

- Q.30 Write short note on entrainment in columns.
 Q.31 Compare absorption and stripping.
 Q.32 Explain selection of solvent for extraction.
 Q.33 Describe Batch distillation.
 Q.34 Explain industrial application of steam distillation.
 Q.35 Describe Bollman's Extraction.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain design and operation of packed absorption column with neat diagram.
 Q.37 Describe binary distillation using McCabe-Thiele method only procedure.
 Q.38 Explain Construction, Working & application of Rotating disc contractor.

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5th Sem / Diploma in Chemical Engineering (P&P) Sub : Mass Transfer II

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M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The bottom product of distillation column is also known as
 a) Distillate b) Residue
 c) Reflux d) Vapor
- Q.2 The equilibrium relation in gas absorption is generally represented by
 a) Raoult's law b) Henry's law
 c) Dalton's law d) Boyle's law
- Q.3 In leaching, the solvent used is called
 a) Raffinate b) Extract
 c) Leachant d) Distillate
- Q.4 Reflux in distillation is used to
 a) Increase temperature
 b) Improve separation
 c) Reduce pressure
 d) Increase feed rate

- Q.5 Crystallization is a process of
- Liquid liquid separation
 - Solid formation from solution
 - Gas absorption
 - Filtration
- Q.6 In extraction, distribution coefficient is the ratio of
- Solute concentration in feed to solvent
 - Solute concentration in two phase
 - Solvent to feed
 - Gas to liquid ratio
- Q.7 Azeotropic distillation is used when
- Relative volatility is high
 - Components form azeotrope
 - Mixture is ideal
 - Pressure is low
- Q.8 The height equivalent to theoretical plate (HETP) is used in
- Plate column
 - Packed column
 - Absorber only
 - Crystallizer
- Q.9 Loading in packed column occurs before
- Flooding
 - Weeping
 - Channeling
 - Foaming
- Q.10 Leaching is a
- Gas liquid operation
 - Liquid liquid operation
 - Solid liquid operation
 - Solid gas operation

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define distillation
- Q.12 For azeotropic composition relative volatility is equal to _____.
- Q.13 Expand RO.
- Q.14 Explain weeping in tray column.
- Q.15 Define Magma.
- Q.16 State Raoult's law.
- Q.17 Define Reflux.
- Q.18 Define Vapour pressure.
- Q.19 For saturated liquid q is equal to _____.
- Q.20 Write two uses of extraction.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain mechanism of gas absorption.
- Q.22 Explain Miers saturation theory.
- Q.23 Explain construction and working of tray column.
- Q.24 Describe differential distillation.
- Q.25 Explain working of Swenson and Walker Crystallizer.
- Q.26 Write short note on packed tower internals.
- Q.27 Explain the effect of impurities on crystal formation.
- Q.28 Describe ideal stages in counter-current leaching.
- Q.29 Explain construction of Oslo crystallization.

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